CHEYANNE SHARIAT

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EDUCATION

Ph.D., Astrophysics, California Institute of Technology — 2024 -
M.S., Astrophysics, California Institute of Technology — 2024 - 2026
DOE Computational Science Graduate Fellow
Advisor: Kareem El-Badry

B.S., Physics, University of California, Los Angeles — 2021 - 2024
summa cum laude
Tisch Academic Excellence Scholar
Advisor: Smadar Naoz

RESEARCH POSITIONS

Graduate Student Researcher — 2024 -
California Institute of Technology — *Pasadena, CA*

Undergraduate Student Researcher — 2021 - 2024
University of California, Los Angeles — *Los Angeles, CA*

Summer Undergraduate Research Fellow — 2023
NASA Jet Propulsion Laboratory (JPL) — *Pasadena, CA*

Summer Undergraduate Research Fellow — 2022
Caltech High Energy Astrophysics Group — *Pasadena, CA*

High School Researcher — 2019 - 2022
UCLA Neurology and Neurogenetics, Fogel Lab — *Los Angeles, CA*

HONORS & AWARDS

DOE Computational Science Graduate Fellowship	2025 – 2029
NSF Graduate Research Fellowship (offer declined)	2025
Raynor L. Duncombe Student Research Prize, AAS DDA	2025
Joshua and Beth Friedman Fellowship, Caltech	2025
Dean's Prize for Excellence in Research, UCLA	2024
Lindau Nobel Laureate Meetings Fellow, Lindau, Germany	2024
Helen Quinn Award for Undergraduate Research Theory (1st), APS	2023
Summer Undergraduate Research Fellowship, NASA Jet Propulsion Laboratory	2023
Undergraduate Research Fellowship, UCLA	2022 – 2023
Summer Undergraduate Research Fellowship, Caltech	2022
Steve Tisch Academic Excellence Scholarship, UCLA	2021 – 2024

GRANTS

Roman General Investigator Grant, NASA/STScI, PI — 2027 – 2030
Binary Star Demographics in the Galactic Bulge

Roman General Investigator Grant, NASA/STScI, PI — 2027 – 2030
RACS: Roman Astrometric Companion Survey

Roman General Investigator Grant, NASA/STScI, Co-I — 2027 – 2030
RAGHAB: Roman Analysis for a Galactic Hunt for Astrometric Binaries

JWST General Observer Grant, NASA/STScI, PI — 2026 – 2029
Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87

TALKS

Stellar Thinkshop, Carnegie Observatories, Pasadena, CA	2026
Astro Coffee, IAS, Princeton, NJ	2026
Thursday Lunch Talk, Princeton, Princeton, NJ	2026
Stellar-Mass Black Holes Workshop, Talk, KITP	2025
The Lifecycle of Stellar Black Holes Conference, Talk, KITP	2025
Carnegie ArXiv Tea, Pasadena, CA	2025
Keck Science Meeting, Poster, UCLA	2025
Binary Stars in the Space Era, Keele, United Kingdom	2025
Vasto Accretion Meeting, Vasto, Italy	2025
AAS Division on Dynamical Astronomy Meeting 56, Atlanta, GA	2025
Undergraduate Research Week, UCLA	2024
Exoplanet Monthly Meeting, Northwestern, Evanston, IL	2024
APS Conference for Undergraduate Women in Physics, San Diego, CA	2024
APS Far West Section, UCSD (Helen Quinn Award, 1st)	2023
Division 3262 Seminar, NASA/JPL-Caltech	2023
Undergraduate Research Week, UCLA	2023
SURF Final Presentation, Caltech	2022
HEAG Presentation Week, Caltech	2022

AWARDED TELESCOPE TIME

James Webb Space Telescope	24.8 hours (PI)
Palomar Hale 200 inch, NGPS	8 nights (PI) 5 nights (Co-I)
Palomar Hale 200 inch, PHARO	9 nights (PI)
Keck HIRESb	9 nights (Co-I)
Keck KCWI	1 night (Co-I)

OBSERVING EXPERIENCE

Palomar Hale Telescope, PHARO – 8 nights	2024-
Keck, HIRES – 2 nights	2025-

MENTORING

Joe Smith (Caltech Undergrad)	2025-
Noah East (UCLA Undergrad)	2025-
Omri Ratzkoff (UCLA Undergrad)	2025-

JOURNAL REFEREE

ApJ, ApJL, MNRAS, A&A, A&AL (*14 papers*) 2023 -

COLLABORATIONS

Sloan Digital Sky Survey 2026 -

LISA Consortium 2025 -

Euclid Consortium 2025 -

TEACHING & OUTREACH

Graduate Teaching Assistant (Caltech) 2025
Ay 101: Physics of Stellar Interiors and Atmospheres

Graduate Peer Mentor (Caltech) 2025 -

LAUSD Guest Speaker 2023 -
James Monroe High School
Shenendoah Early Education Center

Stargazing Lectures (Caltech) 2023 -
Q&A Panelist
Telescope Operator

Diversity, Equity, and Inclusion (DEI) Committee 2023 - 2024
Undergraduate Representative, UCLA Physics and Astronomy Dept. *Los Angeles, CA*

Undergraduate Learning Assistant 2022
Physics 1C, Electrodynamics, Optics, and Special Relativity, UCLA *Los Angeles, CA*

PUBLICATIONS (19 TOTAL; 12 FIRST-AUTHOR)

19. Ding, K., Gennaro, M., Avila, R. J., Ricotti, M., Beaton, R. L., Boyer, M. L., Brown, T. M., Calamida, A., Cassisi, S., Chandra, V., Cohen, R. E., Correnti, M., Crnojević, D., El-Badry, K., Geha, M., Guhathakurta, P., Kallivayalil, N., Kirby, E. N., McQuinn, K. B. W., Savino, A., **Shariat, C.**, Simon, J. D., and Weisz, D. R., "Probing the IMF in the Early Universe – Direct measurements in the Boötes I UFD with JWST/NIRCam", The Astrophysical Journal, *submitted*, arXiv:2605.15069, 2026.

18. **Shariat, C.**, El-Badry, K., MacLeod, M., and Leiner, E., "Gaia astrometry disfavors a binary origin for long secondary periods", arXiv:2604.09767, Publications of the Astronomical Society of the Pacific, 2026.

17. **Shariat, C.**, "OverCite: Add citations in LaTeX without leaving the editor", arXiv:2604.15366, Research Notes of the American Astronomical Society, 2026.

16. Bhattacharjee, S., El-Badry, K., Fuller, J., **Shariat, C.**, Yamaguchi, N., "Thermally inflated accretors in post-mass transfer binaries: Abell 35 and its class revisited", arXiv:2603.23756, Publications of the Astronomical Society of the Pacific, 2026.

15. **Shariat, C.** and El-Badry, K., "A global view of post-interaction white dwarf-main sequence binaries", Publications of the Astronomical Society of the Pacific, vol. 138, no. 3, Art. no. 034202, 2026.

14. **Shariat, C.**, Ye, C. S., Naoz, S., and Rose, S., "Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-Induced Channels", The Astrophysical Journal Letters, vol. 1000, no. 1, Art. no. L17, 2026.

13. **Shariat, C.**, El-Badry, K., Bhattacharjee, S., "How precisely can we measure the ages of subgiant and giant stars?", arXiv:2510.08675, The Open Journal of Astrophysics, 2026.

12. Holzknecht, L., Naoz, S., **Shariat, C.**, "Dynamical Pathways to the Misalignment of the VHS 1256-1257 System", The Astrophysical Journal, vol. 998, no. 1, Art. no. 122, 2026.

11. **Shariat, C.**, El-Badry, K., Gennaro, M., Ding, K., Simon, J.D., Avila, R.J., Calamida, A., Cassisi, S., Correnti, M., Weisz, D.R., Geha, M., Kirby, E.N., Brown, T.M., Ricotti, M., McQuinn, K.B.W., Kallivayalil, N., Gilbert, K., Pacifici, C., Guhathakurta, P., Crnojevi'c, D., Boyer, M.L., Beaton, R.L., Chandra, V., Cohen, R.E., Renzini, A., Savino, A., Tollerud, E.J., "Wide binaries in an ultra-faint dwarf galaxy: discovery, population modeling, and a nail in the coffin of primordial black hole dark matter", Publications of the Astronomical Society of the Pacific, vol. 137, no. 10, Art. no. 104103, 2025.

10. Xuan, Z., **Shariat, C.**, Naoz, S., "From Wide Triples to UCXBs: Multimessenger Detection of Dynamically-formed Black Hole-White Dwarf Systems with LISA", The Astrophysical Journal, vol. 995, no. 1, Art. no. 27, 2025.

9. **Shariat, C.**, El-Badry, K., Naoz, S., "10,000 Resolved Triples from Gaia: Empirical Constraints on Triple Star Populations", Publications of the Astronomical Society of the Pacific, vol. 137, no. 9, Art. no. 094201, 2025.

8. **Shariat, C.**, El-Badry, K., Naoz, S., Rodriguez, A.C., van Roestel, J., "Cataclysmic Variables in Triples: Formation Models and New Discoveries", Publications of the Astronomical Society of the Pacific, vol. 137, no. 7, Art. no. 074201, 2025.

7. Matsuno, T., Kemp, A., Tanikawa, A., **Shariat, C.**, El-Badry, K., Dodd, E., Helmi, A., Koch-Hansen, A.J., Yamaguchi, N., Yan, H., "Unevolved Li-rich stars at low metallicity: a possible formation pathway through novae", Astronomy and Astrophysics, vol. 699, Art. no. A171, EDP, 2025.

6. **Shariat, C.**, Naoz, S., El-Badry, K., Akira Rocha, K., Kalogera, V., Stephan, A.P., Burdge, K., Angelo, I., "Triple Evolution Pathways to Black Hole Low-Mass X-ray Binaries: Insights from V404 Cygni", The Astrophysical Journal, vol. 983, no. 2, 2025.

5. **Shariat C.**, Naoz, S., El-Badry, K., Rodriguez, A.C., Hansen, B. M. S., Angelo, I., Stephan, A. P., "Once a Triple, Not Always a Triple: The Evolution of Hierarchical Triples that Yield Merged Inner Binaries," The Astrophysical Journal, vol. 978, no. 1, 2025.

4. Stephan, A. P., Martin, D. V., Naoz, S., Hughes, N. R., and **Shariat, C.**, "Two Novel Hot Jupiter Formation Pathways: How White Dwarf Kicks Shape the Hot Jupiter Population", The Astrophysical Journal Letters, 2024

3. **Shariat C.**, Hasegawa, Y., Yu, T. Y. M., Hansen, B. M. S., Hu, R., "Predicting the Dominant Formation Mechanism of Multi-Planetary Systems," The Astrophysical Journal Letters, vol. 964, no. 1, 2024.

2. **Shariat, C.**, Naoz, S., Hansen, B. M. S., Angelo, I., Michaely, E., Stephan, A. P., "Dynamical Evolution of White Dwarfs in Triples in the Era of Gaia," The Astrophysical Journal Letters, vol. 955, no. 1, 2023.

1. Lazzarini, M., Hinton, K., **Shariat, C.**, Williams, B.F., Garofali, K., Dalcanton, J.J., Durbin, M., Antoniou, V., Binder, B., Eracleous, M., Vulic, N., Yang, J., Wik, D., Gasca, A., Kuauhtzin, Q., "Multiwavelength Characterization of the High-mass X-Ray Binary Population of M33", The Astrophysical Journal, vol. 952, no. 2, 2023.